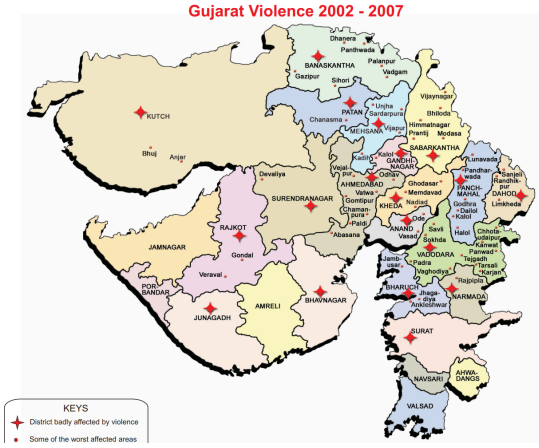
Raja Chinnakotla – Second Draft

I have made significant edits after my first draft. One of the biggest changes with my paper is with my dependent variable, and changing from deaths to buildings destroyed. The reason I changed this variable is that I had an extremely hard time finding reliable and consistent data for deaths from the riot. An instance where I faced the greatest challenge was with a document from Citizens of Justice and Peace (CJP). In a document that cataloged the deaths in districts, the paper became blank, which gave a signal that perhaps there are other avenues that I need to pursue to answer my question. In the draft, I will discuss important literature related to my question, beginning with models and graphs from my cleaned data, and explore anticipated problems that should be addressed by the next draft.

Literature Review

The literature on intercommunal riots between Muslims and Hindus in India focuses on the economic and political factors for why violence exists. A paper by Anirban Mitra and Debraj Ray explores the economic motivations of riots in India before 2002. The paper explains how income variation among Hindus and Muslims is a contributing factor to the intensity of riots. Specifically, they find that increased expenditures from Muslims create an environment more likely for violence to occur. The paper finds that when there is a perceived increase in wealth among Muslim households, Hindu extremists are inclined to destroy the wealth created. The implication is that in the majority of Hindu districts across time periods will respond with violence to positive economic changes among Muslim communities. This paper provides a comprehensive understandnig on motivations for violence among the majority Hindu communities; however, the time series doesn't include periods after 2000, which is relevant for explaining intercommunal violence in Gujarat. Moreover, the paper looks across all regions of India, which presents problems of heterogeneity since provinces across India are vastly different and have other factors that contribute to violence. A paper looking at contributing factors to violence in the 2002 Gujarat riots is by Saumrita Jha. In his analysis, he focused on historical data from Muslim ports and leaders. His theory is that in districts and cities with a history of Muslim leaders, there is greater religious harmony in the region. In his model, he considers factors of distance to Godhra, the BJP vote share, and other factors that contribute to violence. In his paper, he found that towns with medieval ports and that were close to Godhra had statistically significant levels of violence. Jha's paper is useful because of the use of historical data as a contributor to violence. Another paper that looks at the political theory behind the riots is "The Role of the Media During Communal Riots in India: A Study of the 1984 Sikh Riots and the 2002 Gujarat Riots" by Saifuddin Ahmend from NTU. His paper explores the theory of media, particularly regional language papers, in creating non-object media. In the paper, Ahmend points to the Gujarat Samachar and

Sandesh, the largest Gujarati language newspapers, as a perpetrator in insighting violence. One report I found particularly interesting, and will mention later, was a map by CJP mapping all religious monuments destroyed and parsing through which type.



Method Discussion

$$Y_i = \alpha + \beta X_{1i} + \lambda X_{2i} + \tau XD_i + \varepsilon$$

- β – Percentage of Gujarati Mother Tongue Speakers
- λ – Difference between the percentage of Muslims and Hindus in a District
- τ – Binary Variable where 0 = Samachar not printed, 1 = Samachar printed

The method I want to use for my research question is multiple linear regression using a binary variable. I want to use a fairly simple model. In this regression equation, I decided to limit the number of variables because with a maximum of 30 observations, I am limiting the number of independent variables I will include in my regression. I chose these variables because I want to show a causal relationship between language distribution in a community and the amount of monuments destroyed. The decision to use monuments destroyed as my dependent variable came from the difficulty in finding the data for riot outcomes. While the literature uses different metrics, including deaths, vandalism, and police encounters. I found that the easiest data to find on riots at the district level came from Muslim monuments, like Sufi Tombs called Darghas, Masjids, and Islamic schools. This led to the graph below, which I just didn't seem to show a relationship between language distribution and monuments destroyed across the district level. I am starting with a very basic equation because I have not finished compiling and cleaning my datasets with religion. Nonetheless, the model put forth provides good feedback on the viability of the question. Once I am able to synthesize all my data sets, my model will be expanded to the equation I listed above.

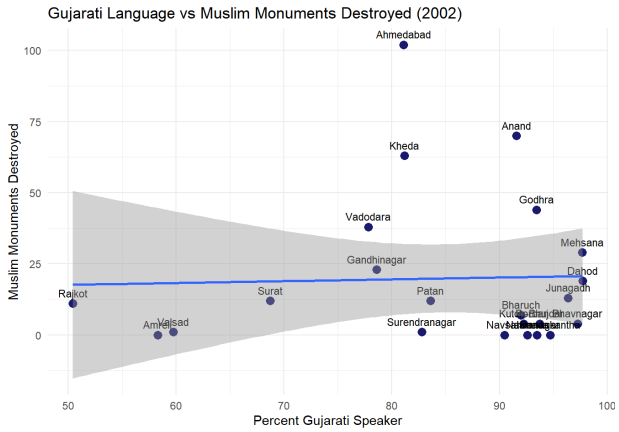


Table 1: OLS Regression: Total Muslim Destroyed

	Estimate	Std. Error	t value	Pr(> |t|)
(Intercept)	-164.6718	133.4827	-1.234	0.232
Pct Gujarat as Mother Tongue	0.3156	0.4590	0.688	0.500
log(Population per Person)	19.7592	14.1757	1.394	0.179
Residual standard error: 26.97 on 20 degrees of freedom				
Multiple R-squared: 0.0895, Adjusted R-squared: -0.001546				
F-statistic: 0.983 on 2 and 20 DF, p-value: 0.3915				

Problems

The resulting table from my simple equation presents many problems. For one, there are no statistically or economically significant variables. The interpretation of the "percent of Gujarati as Mother Tongue" is that a one percent increase in Gujarati language as a mother tongue in a district creates a .32 building destruction. This means nothing since the standard deviation includes both negative and positive values. One problem I need to address is the importance of cities in a district. For instance, Ahmedabad is the largest city in Gujarat, so there is a greater likelihood of riots occurring since many travel to large cities to commit crimes. Moreover, areas with distinct importance attract attention. I plan on addressing this by breaking up districts with large cities as their own district. For instance, classify Ahmedabad City and Ahmedabad Rural as separate districts. I plan to further explore datasets that provide insights into reactions to riots. One set I am looking at is IndiaPoliceEvents, which is headlines compiled from a timeframe in 2002. Looking at my feedback from draft 1, I will make sure to look at other examples with other countries, my reviewer recommended in South America or Africa. Furthermore, I need to provide better justifications for using language as a means for media distribution. I hope to find more relevant literature to support the theories behind regional language and the propaganda of the media. I am extremely excited that I have at least some (albeit not significant) results that are helping with my research forward. Looking forward to where this project takes me!

Citations:

Ahmed, Saifuddin. 2010. “The Role of the Media During Communal Riots in India: A Study of the 1984 Sikh Riots and the 2002 Gujarat Riots.” *Media Asia* 37, no. 2: 103–112. SSRN Scholarly Paper ID 2541129. Social Science Research Network. https://papers.ssrn.com/abstract=2541129.

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Citizens for Justice and Peace. "Gujarat Genocide." CJP India. Accessed March 13, 2026. https://cjp.org.in/gujarat-genocide/.

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Jha, Saumitra. 2013. *“Unfinished Business”: Ethnic Complementarities and the Political Contagion of Peace and Conflict in Gujarat*. NBER Working Paper No. 19203. Cambridge, MA: National Bureau of Economic Research. https://doi.org/10.3386/w19203.